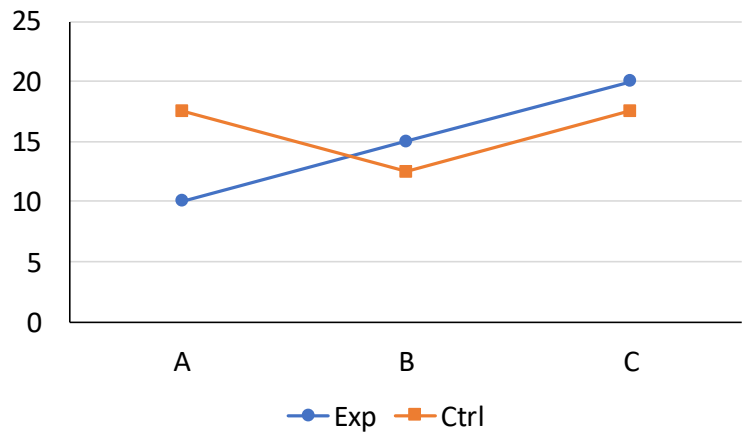
Lab 09A: Multiple Categorical Predictors.

Question 1. Main-Effects and Interactions.

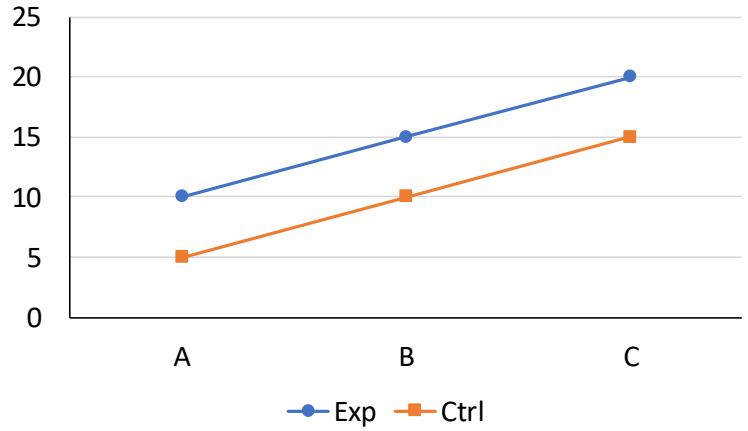
For each of the figures below, indicate if you think there is main-effect of Time (Exp or Ctrl), a main-effect of Condition (A, B, or C), or an interaction of the two factors. Be sure to address all three of these types of effects in your answer and explain your reasoning.

Example 1



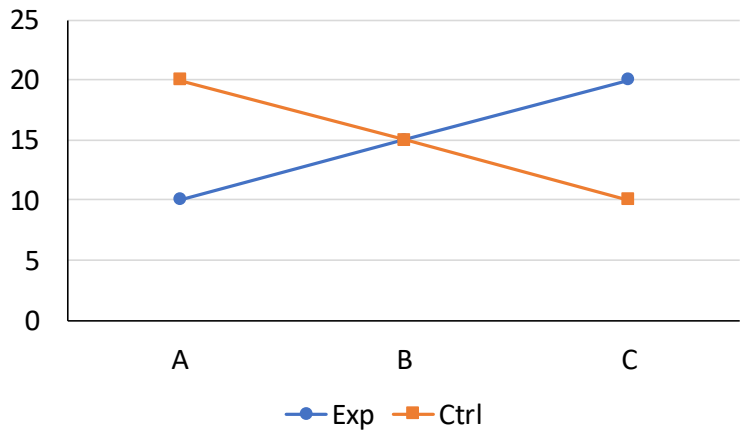
- Your answer:
- **Main Effect of Group**: It does not look like there is a main-effect of group, because (on average across conditions) both groups have an average of about 15.
- **Main Effect of Condition**: There may be a main-effect of condition, because (on average across groups) Conditions A and B have an average of about 14, but Condition C has an average of about 19.
- **Group x Condition Interaction**: There appears to be an interaction, because the effect of Group reverses from Condition A to Conditions B and C.

Example 2



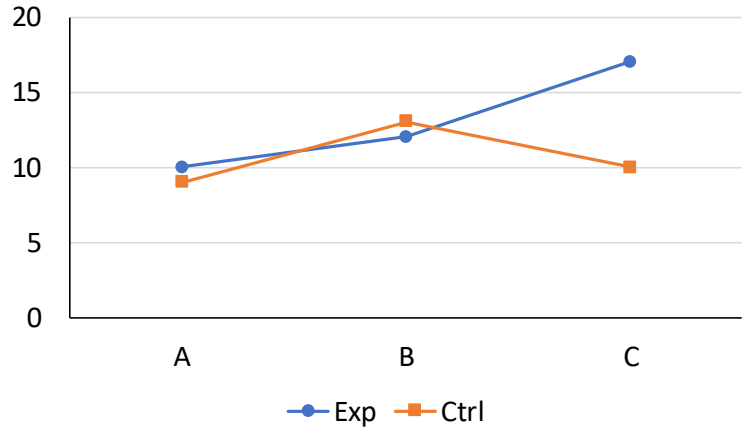
- Your answer:
- **Main-Effect of Group**: There appears to be a main-effect of Group, because (on average across Conditions) the experimental group has a mean of about 15, whereas the control group has a mean of about 10.
- **Main-Effect of Condition**: There appears to be a main-effect of Condition, because (on average across Groups), Condition A has a mean of 7.5, Condition B has a mean of 12.5, and Condition C has a mean of 17.5.
- **Interaction**: There is no evidence of an interaction, because the simple-effect of Group in all conditions is about 5 points.

Example 3



- Your answer:
- **Main-Effect of Group:** It does not look like there is a main-effect of Group, because (on average across conditions) both groups have a mean of about 15.
- **Main-Effect of Condition:** It does not look like there is a main-effect of Condition, because (on average across groups), all conditions have a mean of about 15.
- **Interaction:** There appears to be an interaction, because the effect of Group completely reverses depending on what group you are in.

Example 4



- Your answer:
- **Main-Effect of Group:** It does not look like there is a main-effect of Group, because (on average across conditions) both groups have a mean of about 12.5.
- **Main-Effect of Condition:** It looks like there might be a main-effect of Condition, because (on average across groups), Condition A looks a lot lower than conditions B and C. Condition C also appears to be slightly higher than Condition B.
- **Interaction:** There appears to be an interaction, because the effect of Group is much larger in Condition C, than it is in Conditions A or B.

Question 2. Practice Building Models:

- An educational psychology experiment with a factor of Grade (4th, 5th, and 6th) and Intervention (Experiential Learning versus Traditional Classroom).
  - Write the augmented model including the two main-effects and their interaction:
    - $\hat{y}_i = \beta_0 + \beta_1(G1) + \beta_2(G2) + \beta_3(I1) + \beta_5(G1xI1) + \beta_6(G2xI1)$
  - Write the compact model that you would need to use to calculate **<u>the Omnibus F for the full model</u>**:
    - $\hat{y}_i = \beta_0$

- An exercise intervention with a factor of Group (Low, Medium, and High intensity training) and Diet (Low, Medium, and High protein).
  - Write the augmented model including the two main-effects and their interaction:
    - $\hat{y}_i = \beta_0 + \beta_1(G1) + \beta_2(G2) + \beta_3(D1) + \beta_4(D2) + \beta_5(G1xD1) + \beta_6(G1xD2) + \beta_7(G2xD1) + \beta_8(G2xD2)$
  - Write the compact model that you would need to use to calculate **<u>the Omnibus F for the Group x Diet interaction</u>**:
    - $\hat{y}_i = \beta_0 + \beta_1(G1) + \beta_2(G2) + \beta_3(D1) + \beta_4(D2) + \beta_5(G1xD1) + \beta_6(G1xD2) + 0(G2xD1) + 0(G2xD2)$
    - Or $\hat{y}_i = \beta_0 + \beta_1(G1) + \beta_2(G2) + \beta_3(D1) + \beta_4(D2)$

- A "green" psychology experiment with a factor of Exposure (Real versus Simulated) and Location (Urban, Rural, Forest).
  - Write the augmented model including the two main-effects and their interaction:
    - $\hat{y}_i = \beta_0 + \beta_1(E1) + \beta_2(L1) + \beta_3(L2) + \beta_4(E1xL1) + \beta_5(E1xL2)$
  - Write the compact model that you would need to use to calculate **<u>the Omnibus F for the Main-Effect of Location</u>**:
    - $\hat{y}_i = \beta_0 + \beta_1(E1) + 0(L1) + 0(L2) + \beta_4(E1xL1) + \beta_5(E1xL2)$
    - $Or\ \hat{y}_i = \beta_0 + \beta_1(E1) + \beta_4(E1xL1) + \beta_5(E1xL2)$

- For the green psychology experiment described above, complete the appropriate set of orthogonal contrast codes in the table below:

	Real			Simulated		
Contrast	**Urban**	**Rural**	**Forest**	**Urban**	**Rural**	**Forest**
λ1 (Exp)	-1	-1	-1	+1	+1	+1
λ2 (Loc1)	-1	0	+1	-1	0	+1
λ3 (Loc2)	-1	2	-1	-1	2	-1
λ4 (Exp * L1)	-1*-1 = +1	-1*0 = 0	-1*+1 = -1	+1*-1 = -1	+1*0 = 0	+1*+1 = +1
λ5 (Exp * L2)	-1*-1 = +1	-1*2 = -2	-1*-1 = +1	+1*-1 = -1	+1*2 = 2	+1*-1 = -1

Question 3. Predictions from a Categorical Model:

- Assume that you have a fitted regression model with the following parameters:

$$10.5 - 0.1(drug) - 0.6(exc) - 0.2(drug * exc)$$

This model was fitted based on 40 participants who either received a drug (+1) or not (-1) and either received an exercise regimen (+1) or not (-1), thus creating four different groups with 10 participants in each. Given that these variables are contrast coded and there are equal numbers of participants in each group, calculate:

- The mean of all the participants =
  - 10.5

- The mean of the Drug-Exercise group =
  - $10.5 - 0.1(1) - 0.6(1) - 0.2(1 * 1) = 9.6 = = \hat{y}_{drug-exercise}$

- The mean of the Drug-No Exercise group =
  - $10.5 - 0.1(1) - 0.6(-1) - 0.2(1 * -1) = 11.2 = \hat{y}_{drug-noexercise}$

- The mean of the No Drug-Exercise group =
  - $10.5 - 0.1(-1) - 0.6(+1) - 0.2(-1 * +1) = 10.2 = \hat{y}_{nodrug-exercise}$

- The mean of the No Drug-No Exercise group =
  - $10.5 - 0.1(-1) - 0.6(-1) - 0.2(-1 * -1) = 11 = \hat{y}_{nodrug-exercise}$

- Assume that the interaction above is statistically significant. How would you interpret this result? (In your answer, be sure to address what ***the effect of exercise*** is for people who ***received the drug***, and what the effect of exercise is for people who ***did not receive the drug***.)
- Among people who took the drug, exercise led to a 1.6 point decrease relative to no exercise; whereas among people who did not take the drug, exercise only led to a 0.8 decrease relative to no exercise. As such, the effect of exercise appears to be augmented by the drug.

Question 3. Completing an ANOVA Table

Complete the following ANOVA Table given the Sum of Squared Errors and degrees of freedom provided. These results summarize the results of an experiment in which participants with depression were assigned to receive either a Drug A, Drug B, or a Placebo, and either Psychotherapy or not:

Source	SSE	df	MSE	F_obs	PRE
Reduced (Full Model)	960	5	192.0	46.1	0.95
Drug	453	2	226.5	53.9	0.90
-- X1	441	1	441	105.8	0.90
-- X2	12	1	12	2.9	0.19
Psychotherapy	450	1	450	108.0	0.90
-- X3	450	1	450	108.0	0.90
Drug x Psychotherapy	57	2	28.5	6.8	0.53
-- X4 (X1 * X3)	9	1	9	2.2	0.15
-- X5 (X2 * X3)	48	1	48	11.5	0.49
Error (Model Residuals)	50	12	4.2		
Total	1010	17			

* Note that you can use Figure 9.7 in the book to check your answers, but try to do all of the calculations on your own first!

Complete the following ANOVA Table given the Sum of Squared Errors provided. These results correspond to the exercise science study described above. People were randomly assigned to different exercise Groups (Low, Medium, and High intensity training) and Diets (Low, Medium, and High protein). Note that an equal number of participants (n = 20) were assigned to each group (N = 180):

Source	SSE	df	MSE	F_obs	PRE
Reduced (Full Model)	2203	8	275.375	22.35	0.51
Group	319	2	159.5	12.94	0.13
-- X1	192	1	192	15.58	0.08
-- X2	127	1	127	10.31	0.06
Diet	205	2	102.5	8.32	0.09
-- X3	32	1	32	2.60	0.01
-- X4	173	1	173	14.04	0.08
Group x Diet	1679	4	419.75	34.07	0.44
-- X5	641	1	641	52.02	0.23
-- X6	563	1	563	45.69	0.21
-- X7	10	1	10	0.81	0.00
-- X8	465	1	465	37.74	0.18
Error (Model Residuals)	2107	171	12.32164		
Total	4310	179			